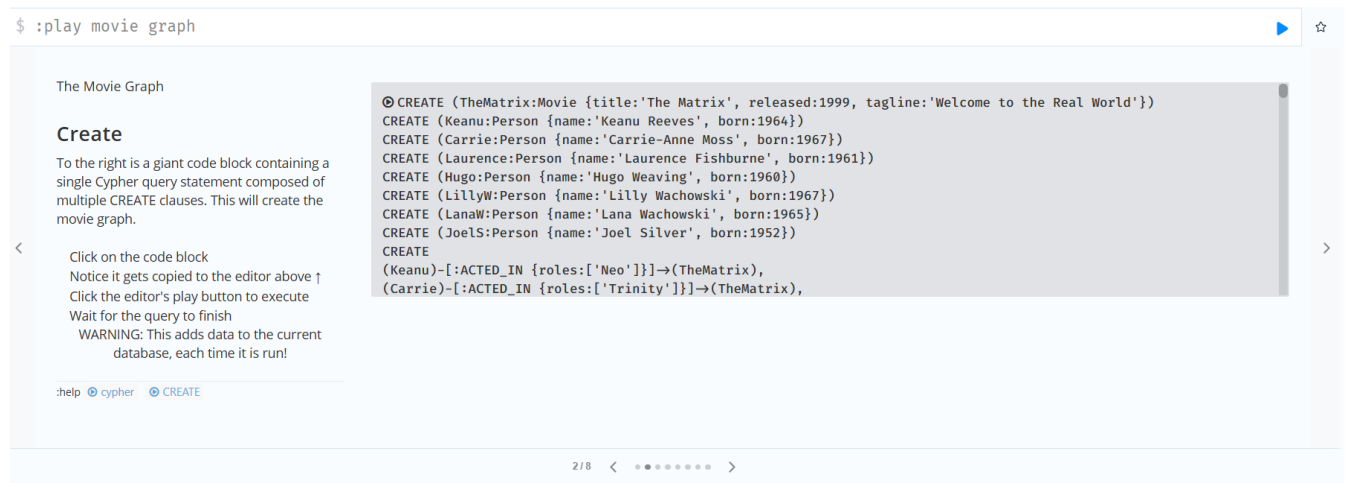


Neo4j Cypher Assignment 2025

Part 1 (Neo4J CRUD Movie Database)

Import the movie database:

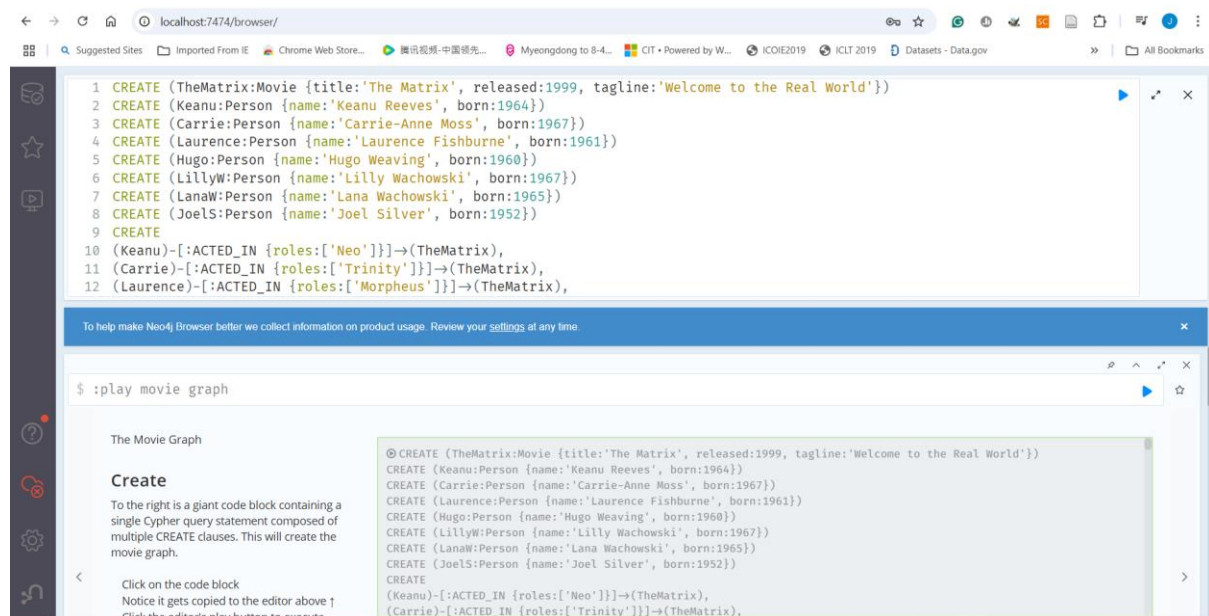
Type in the `:play movie graph` command in the shell panel.



The screenshot shows the Neo4j Browser interface. At the top, a shell panel contains the command `$ :play movie graph`. Below this, the 'The Movie Graph' section is visible. On the left, there's a 'Create' tab with instructions: 'To the right is a giant code block containing a single Cypher query statement composed of multiple CREATE clauses. This will create the movie graph.' Below this, it says 'Click on the code block', 'Notice it gets copied to the editor above', 'Click the editor's play button to execute', and 'Wait for the query to finish'. A warning message states: 'WARNING: This adds data to the current database, each time it is run!'. At the bottom left of this section are links for 'help', 'cypher', and 'CREATE'. On the right, a large code block contains the following Cypher query:

```
CREATE (TheMatrix:Movie {title:'The Matrix', released:1999, tagline:'Welcome to the Real World'})
CREATE (Keanu:Person {name:'Keanu Reeves', born:1964})
CREATE (Carrie:Person {name:'Carrie-Anne Moss', born:1967})
CREATE (Laurence:Person {name:'Laurence Fishburne', born:1961})
CREATE (Hugo:Person {name:'Hugo Weaving', born:1960})
CREATE (LillyW:Person {name:'Lilly Wachowski', born:1967})
CREATE (LanaW:Person {name:'Lana Wachowski', born:1965})
CREATE (JoelS:Person {name:'Joel Silver', born:1952})
CREATE
(Keanu)-[:ACTED_IN {roles:['Neo']}]>(TheMatrix),
(Carrie)-[:ACTED_IN {roles:['Trinity']}]>(TheMatrix),
(Laurence)-[:ACTED_IN {roles:['Morpheus']}]>(TheMatrix),
```

Go to the second page as you seen on the above screenshot. Click on the code and then execute.



This screenshot shows the Neo4j Browser interface in a web browser. The address bar shows 'localhost:7474/browser/'. The browser's address bar and tabs are visible at the top. The main content area shows the 'The Movie Graph' section. The 'Create' tab is active, displaying the same Cypher query as in the previous screenshot. The query is as follows:

```
1 CREATE (TheMatrix:Movie {title:'The Matrix', released:1999, tagline:'Welcome to the Real World'})
2 CREATE (Keanu:Person {name:'Keanu Reeves', born:1964})
3 CREATE (Carrie:Person {name:'Carrie-Anne Moss', born:1967})
4 CREATE (Laurence:Person {name:'Laurence Fishburne', born:1961})
5 CREATE (Hugo:Person {name:'Hugo Weaving', born:1960})
6 CREATE (LillyW:Person {name:'Lilly Wachowski', born:1967})
7 CREATE (LanaW:Person {name:'Lana Wachowski', born:1965})
8 CREATE (JoelS:Person {name:'Joel Silver', born:1952})
9 CREATE
10 (Keanu)-[:ACTED_IN {roles:['Neo']}]>(TheMatrix),
11 (Carrie)-[:ACTED_IN {roles:['Trinity']}]>(TheMatrix),
12 (Laurence)-[:ACTED_IN {roles:['Morpheus']}]>(TheMatrix),
```

Below the query, there's a blue banner that says 'To help make Neo4j Browser better we collect information on product usage. Review your settings at any time'. At the bottom, the shell panel shows the command `$ :play movie graph`.


```
neo4j@neo4j> match (n) return n;
+-----+
| n |
+-----+
| (:Movie {tagline: "Welcome to the Real World", title: "The Matrix", released: 1999}) |
| (:Person {name: "Keanu Reeves", born: 1964}) |
| (:Person {name: "Carrie-Anne Moss", born: 1967}) |
| (:Person {name: "Laurence Fishburne", born: 1961}) |
| (:Person {name: "Hugo Weaving", born: 1960}) |
| (:Person {name: "Lilly Wachowski", born: 1967}) |
| (:Person {name: "Lana Wachowski", born: 1965}) |
| (:Person {name: "Joel Silver", born: 1952}) |
| (:Person {name: "Emil Eifrem", born: 1978}) |
| (:Movie {tagline: "Free your mind", title: "The Matrix Reloaded", released: 2003}) |
| (:Movie {tagline: "Everything that has a beginning has an end", title: "The Matrix Revolutions", released: 2003}) |
| (:Movie {tagline: "Evil has its winning ways", title: "The Devil's Advocate", released: 1997}) |
```

In this assignment, you can choose to use either cypher-shell or standard web browser to complete the tasks. Make sure you screenshot the commands and the outputs of each task.

1. Add movie, actor (three main characters), director as vertices and ACTED_IN/DIRECTED edges for movie “The Bridges of Madison County” [The Bridges of Madison County \(1995\) - IMDb](#) [PS. Do not insert vertices that already exist in the database, and make sure the database does not contain any dummy nodes]
2. Find all actors that directed a movie they also acted in and return actor and movies
3. Find all actor pairs that acted in multiple movies together
4. Find all movies that Keanu Reeves has acted in and the co-actors that acted in the same movie, returning the movie title and the list of co-actors that Keanu Reeves worked with.
5. Delete the movies that have the lowest rating.

Part 2 (Personalized Learning Database)

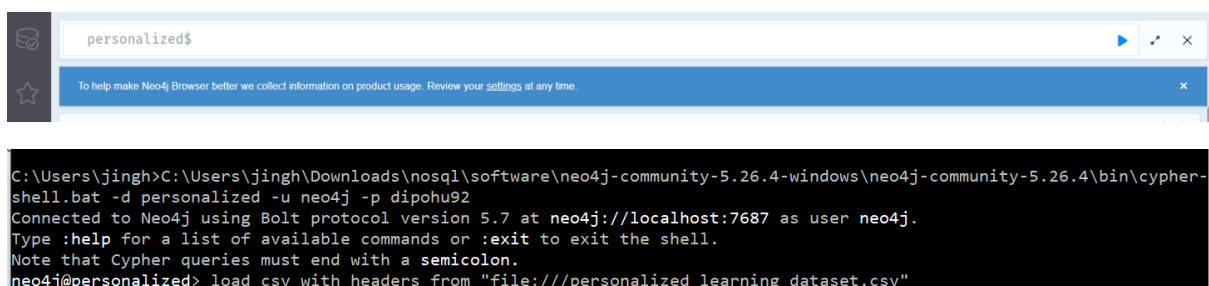
This dataset is designed to support research on **adaptive learning systems**, **personalized education**, and **predictive student success modeling**. It captures rich interaction logs from online education platforms, including student engagement, quiz performance, learning preferences, and dropout likelihood.

Dataset Features

1. **Student_ID** – Unique identifier for each student

2. **Age** – Student's age (15-50 years)
3. **Gender** – The gender of the student (Male, Female, Other)
4. **Education_Level** – The current education level of the student (High School, Undergraduate, or Postgraduate)
5. **Course_Name** – The name of the course the student is enrolled in (e.g., Machine Learning, Python Basics, Data Science, etc.)
6. **Time_Spent_on_Videos (mins)** – The total minutes spent watching educational videos.
7. **Quiz_Attempts** – The number of times a student attempts quiz.
8. **Quiz_Scores (%)** – The percentage score obtained in quizzes.
9. **Forum_Participation (posts)** – The number of forum discussions the student has participated in.
10. **Assignment_Completion_Rate (%)** The percentage of completed assignments by the student.
11. **Engagement_Level** – Low, Medium, High (Based on activity metrics)
12. **Final_Exam_Score (%)** – Percentage score in the final exam
13. **Learning_Style** – Visual, Auditory, Reading/Writing, Kinesthetic
14. **Feedback_Score (1-5)** – Student rating of the course
15. **Dropout_Likelihood (Yes/No)** – Whether the student is likely to drop out

To set up a database named personalized and load the csv data into this database.



```
C:\Users\j\ingh>C:\Users\j\ingh\Downloads\nosql\software\neo4j-community-5.26.4-windows\neo4j-community-5.26.4\bin\cypher-shell.bat -d personalized -u neo4j -p dipohu92
Connected to Neo4j using Bolt protocol version 5.7 at neo4j://localhost:7687 as user neo4j.
Type :help for a list of available commands or :exit to exit the shell.
Note that Cypher queries must end with a semicolon.
neo4j@personalized> load csv with headers from "file:///personalized_learning_dataset.csv"
```

Complete the following queries. Make sure you screenshot the commands and the outputs of each task.

1. List all female young female students, age below 20, prefer to visual learning style on their courses in high school and they are unlikely to drop out their courses.

2. List all courses at various educational level that have been actively engaged by students and have highest assignments completion rate.
3. Which course has received the highest feedback rating score?
4. How many male students have failed the cybersecurity final exam (below 40) due to the low engagement?
5. How many students will likely to drop out their courses due to low engagement and list out their learning style and their educational level.